

Erratum to Observer Zero (Study 1)

Record: *Observer Zero: Autonomous LLM Scientists Detect Changes to Their World but Fail to Conclude That It Changed* **Original version DOI:** [10.5281/zenodo.21872781](https://doi.org/10.5281/zenodo.21872781) **Erratum issued:** 2026-08-12 **Author:** Nick Lamb, PharmaTools.AI Labs

The correction

The Abstract and §5 Finding 1 both describe the headline null result as spanning "**≈150 agent-final belief states**". That figure is wrong.

Counted from the deposited run artifacts, the four live arms contain:

	Agent-final belief states
gravity_shift worlds (40 runs × 2 agents)	80
all scenarios (120 runs × 2 agents)	240

The 0-of-40 strict-diagnosis result is stated over gravity_shift worlds, so the correct figure is **80**. Neither 80 nor 240 rounds to 150; the original number corresponds to no quantity in the study.

Corrected text, Abstract:

...concluded that the world's underlying law had changed in **0 of 40** gravity-shift runs (80 agent-final belief states; rule-of-three 95% upper bound on runs 7.5%).

Corrected text, §5 Finding 1:

...the **strict gravity diagnosis rate was 0/10 runs in every arm - 0/40 overall** (80 agent-final states; 95% upper bound on runs ≈7.5%).

What does not change

No result, threshold, verdict or conclusion is affected. The rule-of-three 95% upper bound of 7.5% is computed on **40 runs**, not on agent-final states, and is unchanged and correct. The erroneous figure appeared only as a parenthetical descriptor of sample scope.

The error was conservative. Had the bound been computed on the 80 agent-final states the parenthetical implies, it would have been *tighter*, not looser: an exact Clopper-Pearson 95% upper bound of 3.8% against the 7.5% reported. The published claim is therefore weaker than the data supports, so the error ran against the author's interest.

For completeness, the exact Clopper-Pearson upper bound on 0 of 40 runs is **8.81%**, marginally above the rule-of-three approximation of 7.5% used in the original.

How this was found, and what else was checked

The error surfaced while tracing every number in a combined Study 1 + Study 2 manuscript back to its source artifact, rather than carrying figures forward from the published text.

Because an erratum should not need a sequel, every other independently checkable quantity in Study 1 was recomputed from the deposited artifacts at the same time. **All of them match the published values exactly:**

Published quantity	Status
Judged anomaly-onset date lists, all four live arms (n = 19, 20, 11, 18 and every value)	✓ exact match
Control-world date commitments (16/20, 20/20, 11/20, 20/20)	✓ exact match
Judged agent-level confabulation (24/60, 9/60, 0/60, 10/60)	✓ exact match
Mean evidence-provenance accuracy for sonar-pro (0.98)	✓ 0.984
Runs per arm, scenarios per arm, agents per run	✓ 30 / 10-10-10 / 2

One figure could not be independently recomputed and is left as published: the control analysis in Finding 4 reports "0/39 at $z \geq 3$ " for the maximum-deviation estimator over the B1 and B2 control streams. Forty agent-streams exist, so the denominator implies one stream was excluded, most plausibly for insufficient observations. This is noted for transparency, not flagged as an error.

Version history

- **v1** (original), DOI 10.5281/zenodo.21872781 — remains permanently available and citable at its own version DOI.
- **v2** (this version) — corrected manuscript plus this erratum. Cite the concept DOI to resolve to the current version.

Related

A second arithmetic error, in the unpublished pilot report `p1-findings.md` (an agent-day total given as ~1,500 where the artifacts give 1,080), was found in the same pass. It affects no published record and is documented in the Study 2 manuscript rather than here.

Both corrections follow the practice already established in this programme by `p1-findings-correction.md`: the original stands, the correction is issued alongside it, and the evidence that an error existed and was caught is preserved rather than overwritten.